CHAIR FOR MOTION PICTURE THEATERS

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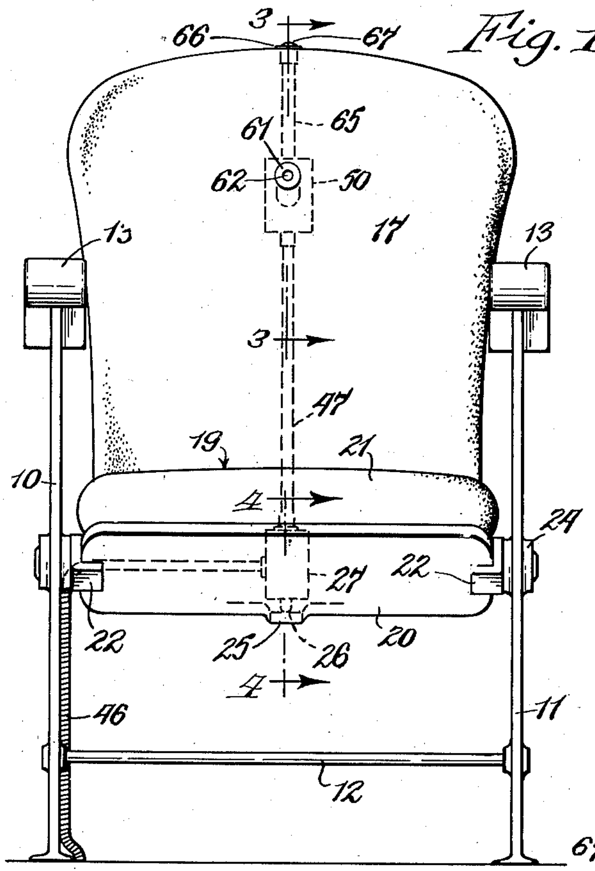


Fig. 1.

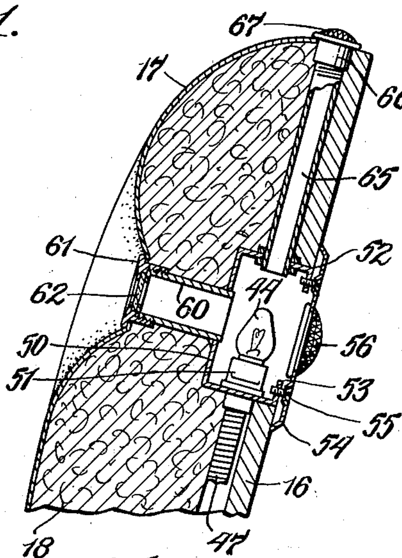


Fig. 3.

Fig. 5.

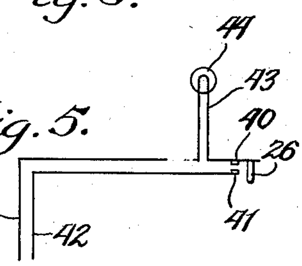
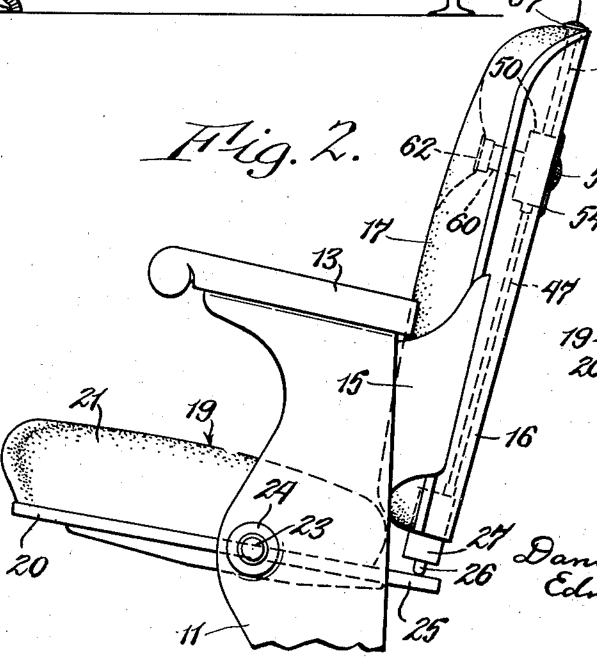


Fig. 2.



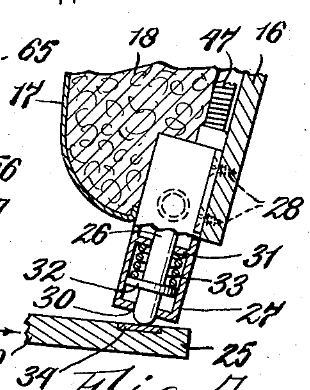


Fig. 4.

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CHAIR FOR MOTION PICTURE THEATERS

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1 Claim. (Cl. 177—311)

This invention relates to a chair for motion picture theaters and more particularly to a seat having an illuminated signal which is lit when the seat is unoccupied so as to facilitate the seating of patrons in the dark theater.

Since the auditorium of motion pictures theaters is dark, considerable difficulty is experienced by patrons in finding seats, particularly when only a few seats are available. It is the principal object of the present invention to provide an individual signal light for each seat which automatically lights when the patron rises and leaves the seat and which is automatically extinguished when the seat is occupied, the signal being visible at a distance so that patrons can quickly find suitable seats. The invention also insures that all seats are occupied when the theater is crowded and also insures the rapid seating of patrons and thereby increases the effective capacity of the house since motion picture theaters generally have continuous performances and each patron usually leaves when the show comes around to the sequence at which the patron had been seated.

Another object of the invention is to provide such an empty seat indicator which is visible at a distance from the seat and from the rear, the light for this purpose being mounted in the back of each seat and above the level of the knees of the seated patrons.

Another object of the invention is to provide such a signal light which can also be constructed to be visible from the front of the empty seat and also from the top thereof, a single bulb being employed to illuminate the rear, front and top signals, and the front signal being so constructed as to not interfere with the comfortable seating of the patron.

Another object of the invention is to provide such an empty seat indicator in which the necessary switch and spring means which cooperate with the swinging seat are mounted on the back of the theater seat, there being no parts which interfere with the movement of the patrons along the rows of seats and there being no parts which interfere with the feet of the seated patrons or with the cleaning of the floor under the seats.

Another purpose of the invention is to provide such an indicator for each seat which will not disturb the patrons even when a large number of seats are empty, it also being understood that when many empty seats are available, the empty seat indicators eacn be rendered inoperative by opening a master switch, since under these circumstances the patrons have no difficulty in finding suitable seats.

Another purpose of the invention is to provide such an empty seat indicator which can be readily adapted to all types of theater seats; which is simple and inexpensive in construction and which consumes only a small amount of electric current.

In the accompanying drawing:

Fig. 1 is a front elevation of a motion picture theater seat equipped with an empty seat indicator made in accordance with my invention.

Fig. 2 is a fragmentary side elevation thereof.

Figs. 3 and 4 are fragmentary vertical sections, taken on the correspondingly numbered lines on Fig. 1.

Fig. 5 is a wiring diagram of the electric circuit included in the empty seat indicator.

The motion picture theater seat can be of any usual and well known construction and is shown as including side standards **10** and **11** which can be made of metal and are usually rigidly secured to the floor, these standards being preferably connected by a cross bar **12** which forms a foot rest for the patrons seated in the rear of the seat and the upper end of each of the standards **10** and **11** carrying arm rests **13**. Each of the side standards carries a rearwardly extending wing or bracket **15**, these brackets **15** jointly supporting the rigid back panel **16** of the seat, this back panel being suitably upholstered by a cloth or leather covering **17** and a substantial depth of resilient padding or stuffing material **18** between the cover **17** and the back panel **16**. The seat **19** comprises a panel **20** which is upholstered, as indicated at **21**, and is supported by side brackets **22** each of which is formed to provide a pin **23**, the pins **23** being journaled in bearings **24** formed in the side standards **10** and **11** so that the seat can be swung from the horizontal position shown to a vertical position and thereby facilitate the movement of patrons along the rows of seats.

The seat panel **20** is also formed to provide a rearwardly extending arm **25** which moves upwardly when the seat is lowered into engagement with a spring loaded plunger **26** mounted on the bottom of the rigid back panel **16**. This plunger is shown as slidingly mounted in a switch box **27** which extends into the upholstery of the back and can be secured to the front side of the back panel **16** by screws **28** or in any other suitable manner. The plunger **26** projects downwardly through an opening **30** in the bottom of the switch box **27** and at its upper end is shown as guided in a slideway provided in an internal cross bar 31 provided in this switch box 27. The plunger is shown as having an integral collar 32 and a relatively heavy compression spring 33 is interposed between the cross bar 31 of the switch box and the collar 32 so as to yieldingly hold the collar 32 in engagement with the bottom of the switch box 27. The end of the plunger is shown as engaging a metal insert 34 in the arm 25 and the spring 33 is of sufficient strength to normally hold the seat 19 in a slightly elevated position.

The plunger 26 operates a switch arranged within the switch housing 27, which switch can be of any suitable construction. As illustrated in Fig. 5 this switch comprises a movable contact 40 and a fixed contact 41, the fixed contact 41 being connected to one side 42 of the power line. The movable contact 40 normally engages the fixed contact 41 and is moved out of engagement therewith by the upward movement of the plunger 26 when the seat 19 is fully depressed by the weight of an occupant. The movable contact 40 is connected by a line 43 with a small electric light bulb 44, the other terminal of this electric light bulb being connected with the other side 45 of the power line. The power lines 45 and 42 are enclosed in a suitable cable 46 leading from the floor along the standard 10 and the front side of the back panel 16 to the switch box 27 and the lines 43 and 45 beyond the switch box are suitably enclosed in a cable 47 which extends along the back panel 16 to a light box 50, the electric bulb 44 being removably mounted in a base 51 mounted on the bottom of this light box.

The light box 50 is mounted in an opening 52 in the back panel 16 near the top of this panel and well above the level of the knees of the sitting patrons. The rear of this light box is open and provided with an inwardly extending flange 53 to which an ornamental cover plate 54 is removably secured in any suitable manner, as by screws 55. The edges of the cover plate 54 can bear against the back panel 16 and this back panel carries a glass jewel 56 mounted in an opening provided in the cover plate 54 immediately in rear of the bulb 44 so that this glass jewel 56 is illuminated when the bulb 44 is lit. The jewel 56 is preferably so faceted as to be clearly visible from any angle when it is illuminated.

A light tube 60 projects forwardly from the front wall of the light box 50 in line with the filament of the bulb 44 and is externally threaded at its front end. An internally threaded collar 61 is secured to the threaded front end of the light tube 60 and carries a glass 62 which can be either in the form of a pane, as shown, or in the form of a jewel. The light tube 60 extends through the upholstery for the back of the seat and to avoid discomfort of the seated patron, the covering 17 is drawn rearwardly and secured to the collar 61, the cover 17 being provided with an opening through which the collar 61 extends and the collar being provided with a peripheral flange for holding the covering 17 in the tufted condition shown. By so drawing in the upholstery and recessing the light tube 60 it will be seen that the collar 61 in no way interferes with the comfort of the occupant of the seat.

A light tube 65 extends upwardly from the light box 50 in line with the bulb 44 and is threaded at its upper end. An internally threaded collar 66 is secured to the upper end of this light tube 65, this collar having a flange which is screwed against the upper edge of the back panel 16. This collar 66 carries a jewel 67 which, through the light tube 65, is illuminated when the bulb 44 is lit and thereby provides an illuminated jewel at the top of the seat back when the seat is unoccupied.

When the seat is unoccupied the compression spring 33 holds the plunger 26 in its downwardly extended position, this plunger bearing against the arm 25 of the seat panel 20 to thereby hold the seat 19 in a slightly elevated position. In this position of the parts the movable contact 40 is in engagement with the fixed contact 41 and hence a circuit is closed through the side 42 of line, fixed contact 41, movable contact 40, wire 43 and bulb 44 to the other side 45 of the line. The bulb 44 is thereby illuminated, the jewels 56 and 67 being thereby illuminated and light also being projected through the light tube 60 to the front of the empty seat. If the patron leaving the seat has lifted the seat 19 to a vertical position it will be seen that the illumination of the bulb 44 is continued since the plunger 26 is held in a depressed position by the helical compression spring 33 and out of engagement with the movable contact 40 so that this movable contact engages the fixed contact 41. When an occupant sits down the seat 19 is depressed slightly against the resistance of the helical compression spring 33, the plunger 26 being moved upwardly against the resistance of this spring by the arm 25. This movement of the plunger 26 moves the movable contact 41 out of engagement with the fixed contact 40, the circuit through the bulb 44 being thereby broken. The illumination of the jewels 56 and 67 and the glass 62 thereby ceases and other patrons are thereby instantly apprised that the seat is occupied, this darkened condition of the indicator continuing as long as the seat is occupied.

It is preferable to provide a sufficient movement of the seat 19 by the helical compression spring 33 that the empty seat indicated is not overly sensitive in its operation. This avoids the indicator being flashed off and on when the seat is occupied by restless children who may move around and effect slight bouncing movements of the seat in their activity.

From the foregoing it will be seen that the present invention provides a very simple and effective illuminated indicator which shows when a seat is unoccupied and which is visible from all sides and for a great distance, thereby facilitating the seating of patrons in a darkened motion picture auditorium. The lighted empty seat indicators at the same time are not disturbing to the seated patrons who are enjoying the show and the indicator in no way interferes with the movement of the patrons in getting to or leaving the seats or with the feet of the seated patrons. It will also be seen that the floor space under the seat is left completely clear so as to avoid any interference with keeping the floor clean and that by locating the jewels above the level of the knees of the seated patrons the indicator clearly shows, from a distance, exactly which seat is unoccupied. It will further be understood that a master switch can be provided in the line supplying power to all of the seats for cutting the indicators into and out of operation, the indicators being unnecessary when a large number of unoccupied seats are available.

It will be understood that substantial alterations can be made in the construction without departing from the invention. For example, the helical compression spring 33 could be exposed and contact directly with the arm **25** instead of through the medium of the plunger **26**. The invention is therefore not to be construed as limited to the particular embodiment shown but is to be accorded the full range or equivalents comprehended by the accompanying claim.

We claim as our invention:

The combination with a chair for motion picture theaters having a back, a seat moved relative to said back by the weight of the patron occupying the chair and a standard supporting said back and seat, of means for indicating when the seat is unoccupied comprising an electric switch mounted on said back and including a part engaged by said seat to open said switch upon the movement of the seat in response to the weight of the occupant, spring means for yieldingly returning said seat to a slightly extended position to close said switch when the occupant arises therefrom, an electric signal lamp mounted on said back directly above said switch and above the level of the knees of the seated patrons for providing an indicator, which is visible at a distance, that the seat is unoccupied, a vertical tube arranged in said back and connecting with said switch and signal and an electric wire in said tube and connecting said signal and switch in circuit with each other.

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